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Q1) Write a function to find the longest common prefix string amongst an array of strings.

If there is no common prefix, return an empty string "".

Example 1:

Input: strs = ["flower","flow","flight"]

Output: "fl"

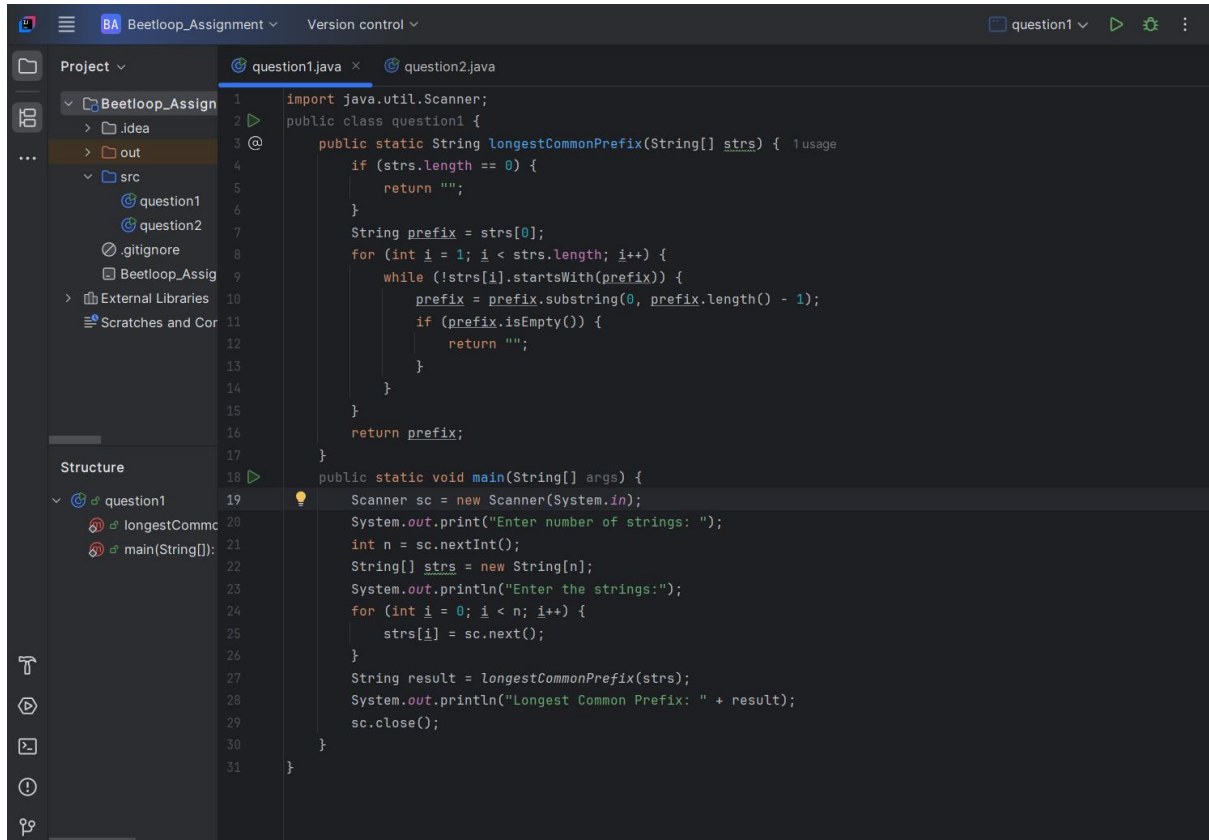
Example 2:

Input: strs = ["dog","racecar","car"]

Output: ""

Explanation: There is no common prefix among the input strings.





```
1 import java.util.Scanner;
2 public class question1 {
3     @
4     public static String longestCommonPrefix(String[] strs) { 1 usage
5         if (strs.length == 0) {
6             return "";
7         }
8         String prefix = strs[0];
9         for (int i = 1; i < strs.length; i++) {
10             while (!strs[i].startsWith(prefix)) {
11                 prefix = prefix.substring(0, prefix.length() - 1);
12                 if (prefix.isEmpty()) {
13                     return "";
14                 }
15             }
16         }
17         return prefix;
18     }
19     public static void main(String[] args) {
20         Scanner sc = new Scanner(System.in);
21         System.out.print("Enter number of strings: ");
22         int n = sc.nextInt();
23         String[] strs = new String[n];
24         System.out.println("Enter the strings:");
25         for (int i = 0; i < n; i++) {
26             strs[i] = sc.next();
27         }
28         String result = longestCommonPrefix(strs);
29         System.out.println("Longest Common Prefix: " + result);
30         sc.close();
31     }
32 }
```

OUTPUT:

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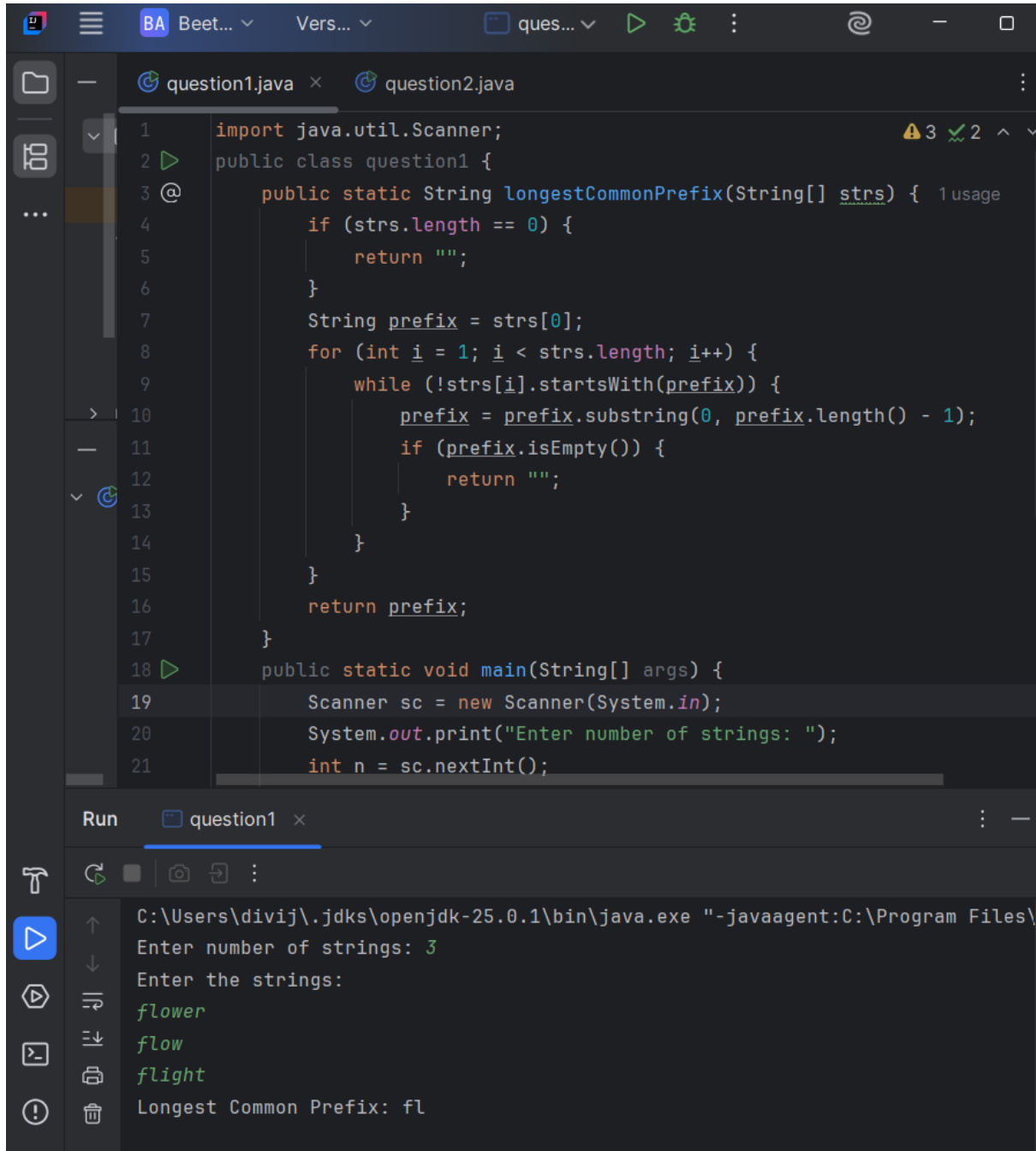
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```
1 import java.util.Scanner;
2 public class question1 {
3     public static String longestCommonPrefix(String[] strs) {
4         if (strs.length == 0) {
5             return "";
6         }
7         String prefix = strs[0];
8         for (int i = 1; i < strs.length; i++) {
9             while (!strs[i].startsWith(prefix)) {
10                 prefix = prefix.substring(0, prefix.length() - 1);
11                 if (prefix.isEmpty()) {
12                     return "";
13                 }
14             }
15         }
16         return prefix;
17     }
18     public static void main(String[] args) {
19         Scanner sc = new Scanner(System.in);
20         System.out.print("Enter number of strings: ");
21         int n = sc.nextInt();
```

Run question1

```
C:\Users\divij\.jdk\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\
Enter number of strings: 3
Enter the strings:
flower
flow
flight
Longest Common Prefix: fl
```

Q2) You are given a **0-indexed** array of integers `nums` of length `n`. You are initially positioned at index 0.

Each element `nums[i]` represents the maximum length of a forward jump from index `i`. In other words, if you are at index `i`, you can jump to any index `(i + j)` where:



- $0 \leq j \leq \text{nums}[i]$ and
- $i + j < n$

Return *the minimum number of jumps to reach index n - 1*. The test cases are generated such that you can reach index n - 1.

Example 1:

Input: nums = [2,3,1,1,4]

Output: 2

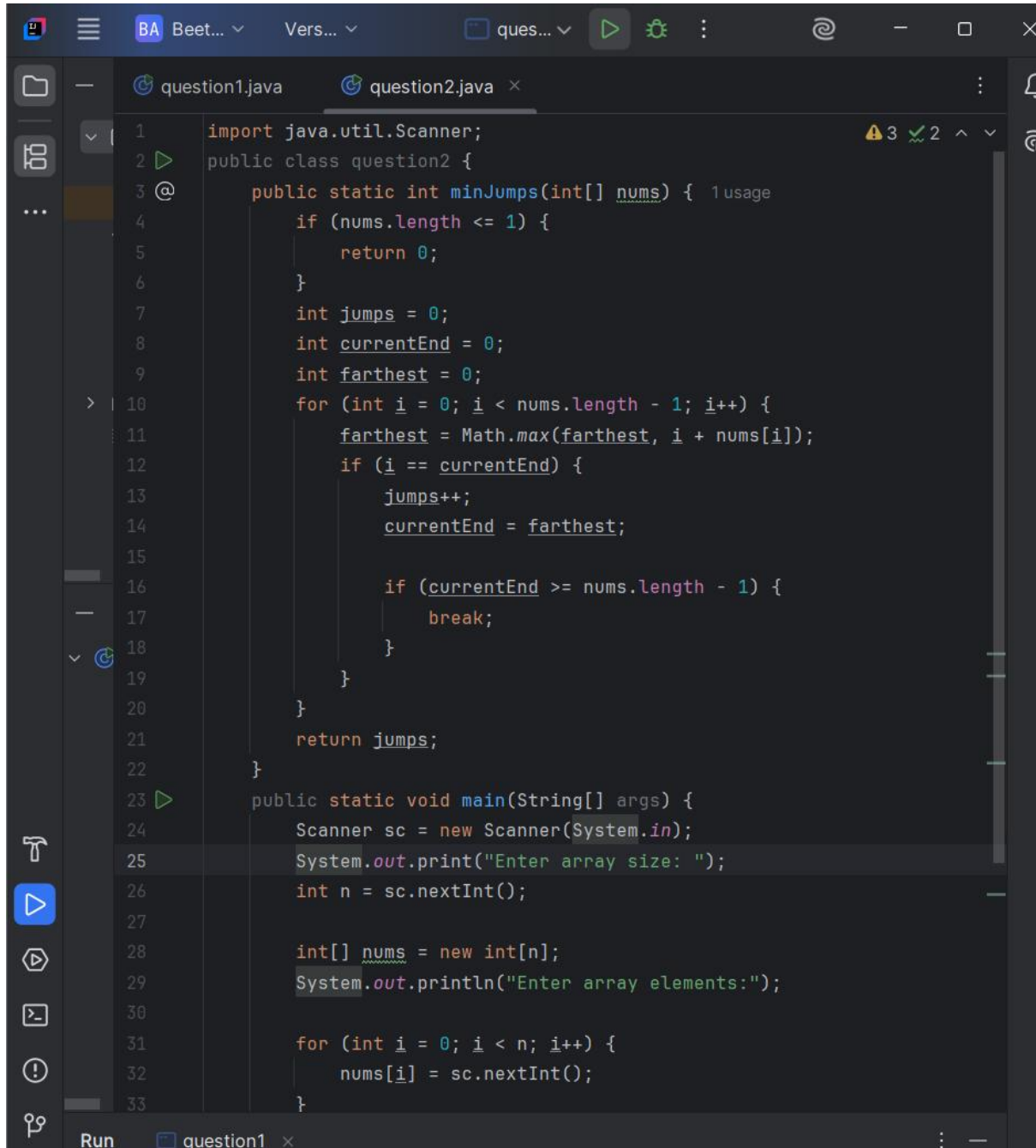
Explanation: The minimum number of jumps to reach the last index is 2. Jump 1 step from index 0 to 1, then 3 steps to the last index.

Example 2:

Input: nums = [2,3,0,1,4]

Output: 2





```
1 import java.util.Scanner;
2 public class question2 {
3     public static int minJumps(int[] nums) { 1 usage
4         if (nums.length <= 1) {
5             return 0;
6         }
7         int jumps = 0;
8         int currentEnd = 0;
9         int farthest = 0;
10        for (int i = 0; i < nums.length - 1; i++) {
11            farthest = Math.max(farthest, i + nums[i]);
12            if (i == currentEnd) {
13                jumps++;
14                currentEnd = farthest;
15
16                if (currentEnd >= nums.length - 1) {
17                    break;
18                }
19            }
20        }
21        return jumps;
22    }
23    public static void main(String[] args) {
24        Scanner sc = new Scanner(System.in);
25        System.out.print("Enter array size: ");
26        int n = sc.nextInt();
27
28        int[] nums = new int[n];
29        System.out.println("Enter array elements:");
30
31        for (int i = 0; i < n; i++) {
32            nums[i] = sc.nextInt();
33        }
```

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```
    }
    return jumps;
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter array size: ");
    int n = sc.nextInt();

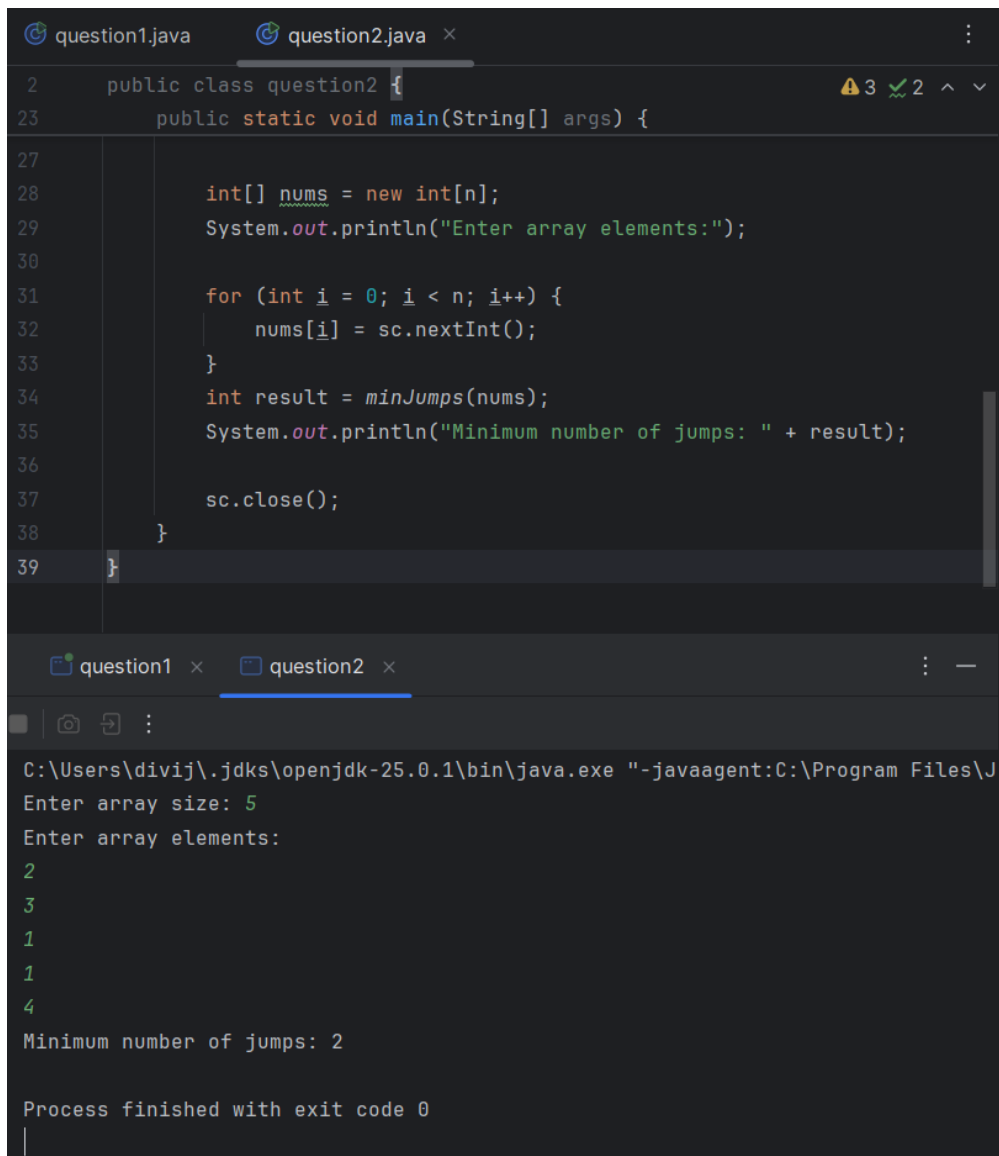
    int[] nums = new int[n];
    System.out.println("Enter array elements:");

    for (int i = 0; i < n; i++) {
        nums[i] = sc.nextInt();
    }
    int result = minJumps(nums);
    System.out.println("Minimum number of jumps: " + result);

    sc.close();
}
```



OUTPUT:



```
2 public class question2 {
23 public static void main(String[] args) {
27
28     int[] nums = new int[n];
29     System.out.println("Enter array elements:");
30
31     for (int i = 0; i < n; i++) {
32         nums[i] = sc.nextInt();
33     }
34     int result = minJumps(nums);
35     System.out.println("Minimum number of jumps: " + result);
36
37     sc.close();
38 }
39 }
```

```
C:\Users\divij\.jdk\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\J
Enter array size: 5
Enter array elements:
2
3
1
1
4
Minimum number of jumps: 2

Process finished with exit code 0
```

